DSP — System Analysis Cheat Sheet

Homework 01 | Linear, Time-Invariant, Causal, Memory

The four tests

Memory	Causality	Linearity	Time-invariance
Does y[n] use only x[n], or also past/future x[n-k] or y[n-k]?	Does y[n] depend on future inputs x[n+k], k>0?	Two sub-tests must both pass:	Two-path test:
Memoryless: only x[n] appears Has memory: x[n-k], x[n+k], or y[n-k] appear	Causal: no future inputs used Non-causal: x[n+1], x[n+2]... appear All memoryless systems are causal.	Scaling: T{a·x} = a·T{x}? Superposition: T{x1+x2} = y1+y2? Red flags: powers, |·|, cos/sin(x), e^x, additive constants	Path 1: replace x[n]→x[n-k], compute output Path 2: take y[n], replace n→n-k Equal → TI. Red flag: explicit n multiplying x.

Q4 — all systems

System	Linear	TI	Causal	Memory
`y[n] = cos(x[n]) + n`	**No**	**No**	**Yes**	None
`y[n] = 2^(-n) * x[n]`	**Yes**	**No**	**Yes**	None
`y[n] = sum(0.9^k * x[n-k])`	**Yes**	**No**	**Yes**	**Yes**
`y[n] = x[n] / (1 + |x[n]|)`	**No**	**Yes**	**Yes**	None
`y[n] = (x[n])^3`	**No**	**Yes**	**Yes**	None
`y[n] = x[n] - x[n-1] + 0.5y[n-1]`	**Yes**	**Yes**	**Yes**	**Yes**
`y[n] = x[n] if |x[n]|<=1, else 0`	**No**	**Yes**	**Yes**	None
`y[n] = e^x[n] - 1`	**No**	**Yes**	**Yes**	None

Quick-fire rules

Explicit n on x	Explicit n coefficient multiplying x[n] → not time-invariant (e.g. 2■n·x[n])
Nonlinear operations	Power, |·|, cos/sin/exp applied to x → not linear
Additive constant	Constant added to output (e.g. +5) → not linear (superposition fails)
Memoryless → causal	A memoryless system is always causal
Sum with upper limit n	Upper limit n on a sum → not TI (limit shifts differently on each test path)
Recursive term	y[n-1] in the equation → has memory, but can still be linear and TI